

EXECUTIVE SUMMARY

Customer Segmentation & Churn Pattern Analytics in European Banking

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1. HEADLINE PERFORMANCE INDICATORS

Total Customers Analysed	Customers Churned
10,000	2,037
Overall Churn Rate	European Industry Benchmark
20.37%	15.00%
Gap vs Benchmark	Total Revenue at Risk
+5.37 pts	EUR 185.6M

83% of revenue risk (EUR 154.4M) is concentrated in High and Premium CLV segments — 1,208 customers representing 59% of all churners. The bank's retention failure is not random; it is concentrated precisely where it is most costly.

2. HYPOTHESIS SCORECARD

REJECTED	H1 — Older customers churn less (switching cost theory) <i>Evidence:</i> Age 46-60 churns at 51.12% — highest of any group, more than double the overall average. Age under 30 churns least at 7.52%. <i>Takeaway:</i> Loyalty does not rise linearly with age. Financially confident mid-life customers are the least loyal, not the most.
CONFIRMED	H2 — Germany churns more due to stronger local competition <i>Evidence:</i> Germany 32.44% vs France 16.15% / Spain 16.67% — almost exactly double. Germany + Age 46-60 reaches 67.33%: the most extreme single segment in the dataset. <i>Takeaway:</i> Country-specific structural problem, not a product-wide issue. Retention strategy must be localised and Germany-specific.
CONFIRMED	H3 — High-balance / high-CLV customers are a hidden high-risk segment

	<i>Evidence:</i> Premium CLV churn 24.20%, High CLV churn 24.12% — both above the 20.37% overall average. Average balances of churned Premium customers: EUR 138,084. <i>Takeaway:</i> The bank disproportionately loses its most profitable customers. Revenue risk is concentrated, not distributed.
PARTIAL	H4 — Inactive members may be dormant savers, not disengaged customers <i>Evidence:</i> Inactive members churn at 26.85% vs Active at 14.27%. BUT zero-balance customers churn least of all balance segments at 13.82%. <i>Takeaway:</i> Two distinct inactive populations exist: genuinely disengaged leavers and quiet zero-balance holders who are not at risk. Segment before intervening.
CONFIRMED	H5 — Female customers churn more <i>Evidence:</i> Female churn 25.07% vs Male 16.46% — a 52% relative gap. Germany + Female reaches 37.55%: highest gender-country combination. <i>Takeaway:</i> Retention messaging and service review should be gender-segmented, not generic.
WEAK	H6 — New customers churn more due to promotional honeymoon behaviour <i>Evidence:</i> New 21.15%, Mid 20.64%, Long 19.51% — direction correct but magnitude minimal. Survival curves confirm no sharp early-tenure churn cliff. <i>Takeaway:</i> Promotional churn may exist but is not a major standalone driver. Likely entangled with product-count effects.

3. KEY ANALYTICAL FINDINGS

A. U-Shaped Product Churn — Most Important Secondary Finding

Number of Products	Churn Rate	Interpretation
1 product	27.71%	Under-engaged — low stickiness
2 products	7.58%	Optimal engagement — safest segment
3 products	82.71%	Over-sold — trust erosion likely
4 products	100.00%	Total loss — every customer churned

2 products = healthy engagement. 3-4 products likely reflects aggressive cross-selling that damages trust rather than building loyalty.

B. Variables That Do NOT Drive Churn

- Credit Score: High 20.18% | Medium 19.64% | Low 22.15% — spread of only 2.5 points. Contradicts the common assumption that creditworthiness predicts loyalty.
- Tenure Group: New 21.15% | Mid 20.64% | Long 19.51% — essentially flat. Loyalty does not meaningfully build over time.

- Credit Card Ownership: Has card 20.18% vs No card 20.81% — statistically indistinguishable. A commonly assumed stickiness feature is irrelevant in this dataset.

C. Survival Analysis — Structural Insights

- Overall survival curve shows steady gradual decline — service quality erosion, not an onboarding shock. H6 (honeymoon churn) not supported.
- Germany separates from France/Spain by Year 2-3 and gap widens continuously. Germany ends at ~35% survival vs ~70% for France/Spain. The problem is structural and requires early-relationship intervention, not just late-stage retention programs.
- Active members maintain ~70% survival at Year 10 vs ~42% for Inactive members. Engagement is a durable protective factor, not just a short-term signal.

D. Feature Correlation Ranking (Point-Biserial)

Rank	Feature	Correlation	Actionable?
1	Age	+0.2853	Yes
2	IsActiveMember	-0.1561	Yes
3	EngagementScore *	-0.1379	Yes
4	Balance	+0.1185	Yes
5	CLV_Score *	+0.1088	Yes
6	NumOfProducts	-0.0478	Yes (non-linear)
7	CreditScore	-0.0271	No (not practical)
11	HasCrCard	-0.0071	No

* Engineered features — EngagementScore and CLV_Score rank above most original variables, validating the value of feature engineering.

4. CUSTOMER SEGMENTATION — K-MEANS CLUSTERING (K=4)						
Cluster	Persona	Customers	Churn	Avg Balance	Status	Priority
0	Active Saver	3,024	15.94%	EUR 122K	Active	Monitor
1	Loyal Everyday Banker	2,127	11.89%	EUR 10K	Active	Benchmark
2	Disengaged Dormant	1,918	20.23%	EUR 7K	Inactive	Low ROI
3	At-Risk Premium	2,931	31.18%	EUR 123K	Inactive	CRITICAL

Critical Insight: Clusters 0 and 3 hold virtually identical average balances (EUR 122K vs EUR 123K) and CLV scores (EUR 93K vs EUR 94K). The sole differentiator is engagement status — Active vs Inactive. Re-engaging Cluster 3 (2,931 customers) is the single highest-ROI retention intervention available to the bank.

5. PREDICTIVE MACHINE LEARNING MODEL

Model Selection — Comparative Evaluation

Model	Recall	Precision	F1-Score	AUC	Selected?
Logistic Regression	0.40	0.39	0.39	0.698	No
Decision Tree	0.64	0.47	0.54	0.826	No
Random Forest	0.64	0.59	0.61	0.851	YES
XGBoost	0.58	0.61	0.59	0.836	No
Tuned Random Forest	0.61	0.63	0.62	--	Alternative

SMOTE (Synthetic Minority Oversampling) was applied to balance the 80/20 class split in training data — growing training samples from 8,000 to 12,740. Test set retained the real-world 20.35% churn distribution for reliable evaluation.

SHAP Explainability — Top Predictors

- Age (SHAP: 0.12) — High age pushes strongly toward churn. Confirms H1 rejection.
- IsActiveMember (SHAP: 0.12) — Inactive status pushes strongly toward churn with near-binaryseparation. Most actionable single feature.
- Geography_Germany (SHAP: 0.088) — Germany residence independently elevates churn risk.
- NumOfProducts — U-shaped SHAP pattern: both very low (1 product) and very high (3-4) push towardchurn. Confirms non-linear relationship.
- Balance (SHAP: 0.022) — Ranks 9th despite being 4th in correlation, confirming that engagement statusmatters more than financial commitment alone.

Risk Scoring — 4-Tier Classification

Risk Tier	Threshold	Customers	Actual Churn Rate	Action
Critical Risk	Prob >= 0.70	232	73.7%	Immediate outreach
High Risk	Prob >= 0.50	213	41.8%	Priority campaign

Medium Risk	Prob ≥ 0.30	410	18.8%	Automated sequence
Low Risk	Prob < 0.30	1,145	6.1%	Monitor only

12x difference in actual churn rate between Critical Risk (73.7%) and Low Risk (6.1%) tiers confirms the model is production-ready for prioritised retention outreach.

6. STRATEGIC RECOMMENDATIONS

1. Germany Retention Task Force

Germany's 32.44% churn is structural and present from Year 2-3 of the customer relationship. A dedicated country-level task force should investigate local competitor offers, service gaps, and branch-level quality. Generic European campaigns will not close this gap.

2. Targeted Re-Engagement for Cluster 3 — At-Risk Premium

2,931 customers hold EUR 123K average balance but are fully inactive. Re-engagement (not financial incentives) is the lever — assign relationship managers, schedule proactive calls, and monitor `IsActiveMember` as the leading indicator. This is the single highest-ROI intervention identified in this analysis.

3. Review Cross-Selling Practices for 3rd and 4th Product Customers

Customers with 3 products churn at 82.71%; with 4 products at 100%. Current cross-selling practices appear to correlate with churn, not loyalty. Review whether product bundling is driven by customer need or sales targets.

4. Gender-Segmented Retention Messaging

Female churn (25.07%) is 52% higher than Male (16.46%) in relative terms. Retention communications, service quality reviews, and relationship manager assignments should account for this gap with tailored approaches.

5. Establish Age 46-60 as a Priority Segment

This group churns at 51.12% — more than double the overall rate. Combined with Germany and female gender, the risk compounds to 67.33%. Proactive outreach before churn signals appear is essential for this cohort.

7. LIMITATIONS AND OPEN QUESTIONS

- Dataset lacks transaction-level, call-centre interaction, and NPS data — root causes (e.g. why Germany churns more) are inferred, not directly observed.
- Low Balance segment churn (34.67%) is based on a small sample (75 customers) — flagged as low-confidence; interpret with caution.
- Product identity (what products 1-4 actually are) is assumed from typical European retail banking patterns, not confirmed by the dataset itself.
- Model recall of 64% means 36% of actual churners are missed. Transaction frequency, login activity, and customer service history would materially improve this if available.
- Churn probability scores are calibrated on SMOTE-balanced training data — raw probabilities may be slightly inflated vs true population base rates.

This executive summary covers a 10,000-customer European banking dataset analysed across descriptive, diagnostic, predictive, and prescriptive analytics stages. Full methodology, statistical tests, and model details are documented in the accompanying research paper and live Streamlit dashboard.

GitHub: github.com/thedhananjaysingh07-dev/BankChurn-Analytics